07 Provenance

Cryptographic Provenance Benchmark — ZULIP

Company: Anticloud FZ LLE

Receipt Generation Performance

Operation	Time
Single receipt creation	0.8ms
Chain verify (100 entries)	2.1ms
Chain verify (10K entries)	198ms
File hash (1MB)	4.2ms
File hash (1GB)	4,200ms

Chain Integrity Tests

Test	Result
Sequential append	PASS
Tamper detection (modify entry 50 of 100)	PASS — detected at entry 50
Reload from disk	PASS — head matches
Cross-node receipt verification	PASS — node fingerprint stable

Hash Algorithm

SHA3-256 (NIST FIPS 202). Collision resistance: 2^128.
No MD5, no SHA1, no SHA2-256 (SHA3 only).